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Assessment of postgraduate digital research literacy: current competencies, desired skills, and enhancement strategies

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Abstract

The rapid development of digital technologies has driven the transformation of scientific research methods and created new requirements for researchers' digital literacy. The cultivation of postgraduate students' digital research literacy is thus of great significance for scientific research. This study aims to understand the current situation, improvement paths, and cultivation needs of digital research literacy among postgraduate students, and to provide suggestions for universities to develop relevant training programs. This study developed an evaluation scale, based on a survey of expert opinions, to assess graduate students' digital research literacy. Online questionnaire data were collected from 314 master's and doctoral students at H University in China. The digital research literacy scale demonstrated strong reliability and validity, as evidenced by Cronbach's alpha coefficient and Confirmatory Factor Analysis (CFA) results. Descriptive statistics and differential analysis were conducted on the survey data. The results show that the overall level of the postgraduate students' digital research literacy was good, but there was still room for improvement. Autonomous learning and self-exploration were the main paths by which students were improving their digital research literacy. Meanwhile, the participants voiced a strong demand for courses and training from the university and for improvement in various dimensions of their digital research literacy. This study suggests that universities should build a targeted and systematic training system for postgraduate students' digital research literacy, including increasing the supply of courses and training, enriching autonomous learning resources, carrying out targeted training based on group characteristics, providing guidance in the actual participation of research projects, strengthening the cultivation of online academic communication skills, and establishing a university-level digital research literacy support team.

Extended author information available on the last page of the article

Keywords Digital literacy · Digital competence · Research literacy · Research competence · Digital research literacy

1 Introduction

Digital literacy is an individual's ability to understand and use digital technology for effective communication, problem-solving, and knowledge creation (Eshet-Alkalai, 2004; Gilster, 1997). Digital literacy has become increasingly important as the rapid development of modern digital technologies impacts various aspects of our learning, work, social interactions, and daily life (Bawden, 2001; OECD, 2019). Major international organizations now place great emphasis on digital literacy education. For example, the European Union listed "Digital Competence" as one of the core competencies for European citizens in 2006 and released the first evaluation framework, DigComp1.0, in 2013 (European Commission, 2021), while UNESCO published "A Global Framework of Reference on Digital Literacy Skills for Indicator 4.4.2" in 2018 (UNESCO UIS, 2018).

The increasing attention to digital literacy has prompted deeper discussions of the topic (Wu et al., 2024). On the one hand, this involves the subdivision of the subjects of digital literacy, as studies have focused on the digital literacy of a range of specific groups, such as teachers (Skantz-Åberg et al., 2022), university students (Techataweewan & Prasertsin, 2018), children (Erstad et al., 2019), and the elderly (Oh et al., 2021). On the other hand, it also involves the subdivision of the scope of digital literacy, with studies on digital literacy in specific tasks or fields such as digital financial literacy (Zaimović et al., 2024), digital health literacy (Patil et al., 2021), digital reading literacy (Chen & Xiao, 2024), etc. The current study focuses on postgraduates as its subject, and digital research literacy as its scope.

The rapid development of digital technology is not only changing the face of education and teaching, but also continuously driving the transformation of research methods and even research paradigms (Tsatsou, 2016; Harrington et al., 2000). The continuous development of informatization and digitalization in scientific research has given rise to tremendous changes in research environments, modes, tools, and methods that are still undergoing constant innovation and change (Brunetti et al., 2023; Crowley et al., 2002; Li et al., 2024). Researchers must now possess the digital literacy essential to scientific research, namely digital research literacy, to enhance the efficiency of their research and the quality of its outcomes (Guillén-Gámez et al., 2023, 2024; Leshchenko et al., 2021).

Scientific research is a primary function of research-oriented universities, and postgraduates are a key workforce in university research. Digital literacy education for postgraduate students should be integrated into their research literacy education (Indah et al., 2022; Li et al., 2020) to better cultivate their digital research literacy. Cultivating digital research literacy can enhance postgraduates' research capabilities and efficiency, leading to more and higher-quality research outcomes (Leshchenko et al., 2021). Beyond the university, it can lay a solid foundation for postgraduates to adapt to the digital society at large and prepare them for future job positions (Robelo & Bucheli, 2018).

However, many universities exhibit an attitude of laissez-faire development regarding the digital research literacy of postgraduates, with targeted actions and plans for improvement being very fragmented, unsystematic, or nonexistent (Bell, 2021; Yan & Fan, 2021). This is highly detrimental to the development of postgraduates' digital research literacy and their adaptation to the research environment of the digital age. Therefore, this study takes H University as a case study and, based on an understanding of the current status and training expectations for postgraduates' digital research literacy at the university, it formulates targeted training strategies and plans.

China is the world's second-largest economy, and its Information and Communication Technology (ICT) industry holds a significant position on the global stage. In the context of China's rapid digital transformation and its emphasis on innovation-driven development, postgraduates' digital research literacy has become a critical factor in advancing academic research and technological innovation. As a leading research-oriented university, H University exemplifies the challenges and opportunities faced by Chinese institutions in cultivating digital research literacy among postgraduates. Its focus on engineering—a field that heavily relies on digital tools and technologies—makes it a particularly relevant case for studying digital research literacy. Studying the digital research literacy of graduate students at H University can offer valuable lessons for other institutions, both domestically and internationally, that aim to enhance their research capabilities.

2 Literature review

Relevant research to date in the field of higher education has examined the digital literacy of both teachers and students (Sillat et al., 2021). In terms of teachers' digital literacy, studies have focused on digital literacy in teaching, which involves questions of how to effectively integrate technology into teaching activities to facilitate student learning and the assessment and selection of appropriate digital tools and resources to support teaching and the evaluation of student learning outcomes (Santos et al., 2021; Pérez-Rivero et al., 2022; Cabero-Almenara et al., 2021; Skantz-Åberg et al., 2022). By contrast, studies on student digital literacy have focused on the technical skills required for personal autonomous learning and lifelong learning, including the ability to find, analyze, and integrate information using information technology, the ability to use digital tools to complete assignments, conduct project research, and collaborate with others, as well as good digital citizenship awareness (Miranda et al., 2018; Techataweewan & Prasertsin, 2018). Relatively speaking, the literature has paid less attention to the digital literacy of university faculty and students from a scientific research perspective (Guillén-Gámez et al., 2023).

2.1 The relationship between digital literacy and research

Overall, digital literacy has a positive facilitative effect on research. Yazon et al. (2019) revealed strong and significant relationships among general digital literacy, digital teaching competence, and research productivity. Friday et al. (2024) found

that librarians' digital literacy had a positive, but non-significant, impact on their research productivity. Guillén-Gámez et al. (2023) found that digital flow and digital skills have a significant impact on university teachers' effective integration of ICT into the research process. Research on students has similarly found a significant correlation between students' digital literacy and research ability (Indah et al., 2022; Liu, 2017; Yang, 2020).

2.2 From research literacy to digital research literacy

Research literacy is demonstrated by an individual's ability to identify research questions during the research process, to systematically apply scientific methods for in-depth investigation and effective resolution of those questions, and to master a range of skills that enable them to successfully achieve the established research objectives (Böttcher & Thiel, 2018; Zhao & Huang, 2024). With the widespread application of information technology today, information literacy and research literacy are often intertwined. For example, "information and research competence"—a combination of information competence and research competence—represents the ability to conduct research based on acquired knowledge, skills, abilities, and experiences; the ability to search for and select digital technologies for the transformation, storage, and transmission of information; the ability to critically evaluate discovered information; and the ability to carry out research using digital technologies (Leshchenko et al., 2021; Liu et al., 2014; Yang, 2015).

Studies on digital literacy in the field of higher education have primarily focused on teaching or learning literacy. Suyo-Vega et al. (2022) proposed a more comprehensive definition of digital literacy that encompasses both teaching and research, arguing that teachers should not only possess digital competencies in teaching but also acquire skills for effective research in the digital environment, including the use of online databases, online surveys, and digital tools for data analysis. Guillén-Gámez et al. (2024) stressed the importance of digital competence for higher education teachers in research work and provided a valuable tool for assessing and improving their digital competence. Tang (2021) introduced the concept of "digital academic ability", and on the basis of prior studies on "digital scholarship" and "academic ability", constructed an evaluation system for digital academic ability. Peng (2022) constructed an index system for the digital academic ability of humanities scholars.

The effective use of digital technologies is crucial for efficiently conducting high-quality research. This study argues that it is necessary to propose a specific form of digital literacy for scientific research (i.e. digital research literacy). We define digital research literacy as the comprehensive literacy, skills and key capabilities required to conduct research activities in an increasingly digitalized environment. It represents the specific manifestation of digital literacy in the context of scientific research and reflects the integration of digital literacy with research literacy.

2.3 Theoretical frameworks in relevant literature

The theoretical frameworks discussed in the relevant literature offer a comprehensive understanding of digital competence, research skills, and information literacy

among higher education teachers and students. These frameworks serve as important references for assessing and enhancing the digital research literacy of individuals in academic settings.

The relevant theoretical frameworks are highly diverse. Many studies employ single-level frameworks. The framework by García Robelo and Veytia Bucheli (2018) encompasses six key skills: inquiry, organizational, analytical, collaborative, communicative, and reflective skills. This model highlights the practical application of ICT tools across various stages of research, ranging from data collection and analysis to collaboration and communication. Similarly, Ipanaqué-Zapata et al. (2023) propose a framework that emphasizes the integration of digital tools throughout the research process. Guillén-Gámez and colleagues have developed a framework for measuring digital competence in research among higher education teachers (Guillén-Gámez et al., 2023, 2024). Their instrument is structured around multiple dimensions, including digital skills, digital ethics, digital flow, anxiety toward ICT use, quality of ICT resources, intention to use ICT, and ICT resource integration. Liu et al. (2014) propose a single-level framework for research information literacy, which includes four key dimensions: information awareness, information retrieval, information evaluation, and information integration.

Some studies adopt dual-level frameworks. Leshchenko et al. (2021) provide a comprehensive framework for assessing information and research competence, particularly in the context of digital transformation. Their model is organized into four criteria: value-motivational, cognitive, activity, and evaluative-reflexive. Each criterion is further divided into multiple indicators, covering a wide range of skills, from motivation and knowledge acquisition to practical application and self-reflection. Yang (2015) offers an evaluation framework structured around four primary dimensions—information awareness, information knowledge, information ability, and information ethics—with twelve secondary indicators. Tang (2021) and Peng (2022) both contribute to the understanding of digital academic ability. Tang's framework includes three primary dimensions: digital information acquisition, digital technology application, and digital academic communication, along with twelve secondary indicators (Tang, 2021). Peng's model, on the other hand, consists of three dimensions: digital academic foundation, digital problem-solving, and digital academic enhancement, along with ten secondary indicators (Peng, 2022). Both frameworks emphasize the critical role of digital tools in enhancing academic research and communication.

The theoretical frameworks in previous literature highlight the multifaceted nature of digital competence and research skills. They underscore the importance of integrating digital tools into research practices while also considering cognitive and ethical factors. These frameworks provide valuable insights into the various dimensions of digital research literacy.

2.4 Postgraduates' digital research literacy

ICT has become an indispensable component of postgraduate students' learning and research paths. Postgraduate students must develop their digital skills in addition to their professional knowledge. Studies to date have emphasized the importance of improving postgraduates' digital research literacy and have provided some strategic

suggestions (Romaniuk et al., 2024; Bell, 2021; Leshchenko et al., 2021; Indah et al., 2022). Romaniuk et al. (2024) studied the use of ICT in doctoral thesis research and concluded that conducting research in the digital age requires that doctoral students possess the appropriate digital research skills. Given that there is still room for improvement in the digital literacy and research literacy of ESL students in Indonesia, Indah et al. (2022) suggested that research skills and digital literacy education should be more effectively integrated into curriculum design. Yan and Fan (2021) argued that the research information literacy education in its current form is divorced from the real scientific research context and an information literacy education system for the whole process of postgraduate scientific research should be constructed instead. Students report that their digital research skills are mainly self-taught, and Bell (2021) concluded that some students need appropriate as-needed training to improve their research productivity. Therefore, it is very important to provide targeted institutional support for postgraduate students.

In summary, digital technology has permeated all aspects of postgraduate research activities, exerting a far-reaching impact that is profoundly transforming learning and research practices. However, studies on digital research literacy remain scarce to date, and there is a lack of understanding regarding the current state, needs, and cultivation of postgraduate digital research literacy. Therefore, this study aims to provide a reference for the development of targeted training programs, so that postgraduate students can better adapt to and excel in learning and research in a digital environment.

3 Research questions

This study aims to provide suggestions for universities to formulate training plans, by answering the following questions:

(1) What is the current state of postgraduates' digital research literacy?
(2) Through which channels do postgraduates improve their digital research literacy?
(3) What are the training needs for postgraduates in terms of digital research literacy?

4 Methods

4.1 Survey sample

A total of 314 postgraduate students from H University, a research-oriented university in China known for its engineering program, participated in the questionnaire survey. Using a convenience sampling technique, an online questionnaire was distributed via Wenjuanxing, a popular online survey platform in China, to collect data. The questionnaire was distributed to the postgraduate student body of various faculties in H University through WeChat, a popular social application. Participants received a link that included an introduction to the study, an assurance of anonymity, and the specific questions of the survey. All respondents participated voluntarily. Among the participants, 188 were male and 126 were female; 206 were master's students and 108 were

doctoral students; and 250 students were from the nature science and engineering fields, while 64 were from the humanities and social sciences.

4.2 Instrument

This study used a questionnaire comprising four parts. The first part collected the participants' background information, such as gender, grade (master's or doctoral), and major field (nature science and engineering or humanities and social sciences). The second part was the Postgraduate Digital Research Literacy Scale (PDRLS), which consists of 30 items divided into seven dimensions: digital research awareness (4 items), digital research knowledge (4 items), literature skills (4 items), auxiliary tool usage (3 items), research data processing (5 items), academic communication (6 items), and digital research ethics (4 items). The development of this scale will be introduced in detail below. The third part addressed the Current Promotion Channels for postgraduate digital research literacy, consisting of 8 items derived from the relevant literature and practical experience. The fourth part addressed the Promotion Needs for postgraduate digital research literacy, encompassing channel needs (8 items, corresponding to the current promotion channels) and content needs (7 items, corresponding to the dimensions of digital research literacy). All items are measured using a five-point Likert scale from 1 = *strongly disagree* to 5 = *strongly agree*.

The Postgraduate Digital Research Literacy Scale used in this study was based on existing research literacy and digital literacy scales (Guillén-Gámez et al., 2023, 2024; Leshchenko et al., 2021; Ipanaqué-Zapata et al., 2023; Yang, 2015; Tang, 2021; Peng, 2022; García Robelo & Veytia Bucheli, 2018), revised in accordance with this study's focus on postgraduates. A list of 53 items related to postgraduate digital research literacy was compiled and divided into seven dimensions. Then, an expert survey questionnaire was created and seven experts were invited to score the importance of the listed items from 1 = *very unimportant* to 5 = *very important*. Based on the expert scoring results, items with an importance score of less than or equal to 4 were deleted. The descriptions of the items were also modified and improved based on the experts' feedback to form the Postgraduate Digital Research Literacy Scale, which ultimately consisted of 30 items. English translations of the specific items of the scale from the Chinese version can be found in Appendix 1. Compared to research literacy more broadly, digital research literacy places a greater emphasis on the ability to use digital tools and technologies in research activities, as well as adhering to the relevant ethics and norms of the digital age and digital environment.

4.3 Data analysis

Data analyses were conducted using SPSS 25.0 and Amos 24.0, namely reliability and validity analysis of the scales, descriptive statistics of the questionnaire items, and difference analysis. The current state of postgraduates' digital research literacy, current promotion channels, and training needs were analyzed using descriptive statistics. Independent sample T-tests were used to analyze the differences between genders, fields of study, and master's versus doctoral students.

5 Results

5.1 Scale reliability

The Cronbach's alpha coefficient was used to measure the reliability of the scale (see Table 1). The overall coefficient of the Postgraduate Digital Research Literacy Scale is 0.940, and the coefficients of all seven dimensions are greater than 0.75, indicating a high degree of scale reliability.

5.2 Scale validity

The scale developed in this study was based on existing frameworks for measuring digital literacy and research literacy. The questionnaire was then adjusted and revised according to expert importance ratings and suggestions, as described above, ensuring the content validity of the questionnaire.

Confirmatory Factor Analysis (CFA) was conducted using Amos 24.0 to test the significance of the items under the scale constructs. The results (see Table 2) show that the standard factor loadings for all items are greater than 0.5, with the majority (70%) of the standard loadings coefficients being greater than 0.7, indicating a significant relationship between the items and their underlying constructs (Hair et al., 2010). The Average Variance Extracted (AVE) and Composite Reliability (CR) were also calculated for all constructs to test the convergent validity and reliability (see Table 2). All AVE values were found to be greater than 0.5, and all CR values were found to be greater than 0.7, indicating a high degree of convergent validity for the model (Fornell & Larcker, 1981; Nunnally & Bernstein, 1994). Discriminant validity analysis was also conducted, and all values indicate acceptable discriminant validity (see Table 3).

According to the criteria for model fit ($X^2/df < 3$, CFI > 0.90, TLI > 0.90, RMSEA < 0.08, and SRMR < 0.08) (Hair et al., 2010; Kline, 2023), the model in this study has an acceptable fit ($X^2/df = 2.319$, CFI = 0.916, TLI = 0.904, RMSEA = 0.061, SRMR = 0.059).

5.3 Descriptive statistics

Table 4 presents the descriptive results for the digital research literacy and desired training content. The mean values of the seven dimensions of postgraduate digital

Table 1 Cronbach's alpha coefficients of the PDRLS

Dimension	Number of Items	Cronbach's α
Digital Research Awareness	4	0.849
Digital Research Knowledge	4	0.903
Literature Skills	4	0.824
Auxiliary Tool Usage	3	0.771
Research Data Processing	5	0.861
Academic Communication	6	0.892
Digital Research Ethics	4	0.879
Overall	30	0.940

Table 2 Results for the measurement model

Factor (Dimension)	Items	Standardized factor loadings	AVE	CR
Digital Research Awareness	Q1	0.569	0.607	0.858
	Q2	0.835		
	Q3	0.896		
	Q4	0.778		
Digital Research Knowledge	Q5	0.822	0.702	0.904
	Q6	0.873		
	Q7	0.840		
	Q8	0.816		
Literature Skills	Q9	0.826	0.555	0.829
	Q10	0.877		
	Q11	0.679		
	Q12	0.553		
Auxiliary Tool Usage	Q13	0.850	0.563	0.789
	Q14	0.812		
	Q15	0.555		
Research Data Processing	Q16	0.812	0.566	0.865
	Q17	0.862		
	Q18	0.782		
	Q19	0.607		
	Q20	0.666		
Academic Communication	Q21	0.613	0.588	0.893
	Q22	0.740		
	Q23	0.762		
	Q24	0.904		
	Q25	0.900		
	Q26	0.627		
Digital Research Ethics	Q27	0.625	0.661	0.884
	Q28	0.757		
	Q29	0.936		
	Q30	0.897		

Table 3 Discriminant validity analysis

	Factor1	Factor2	Factor3	Factor4	Factor5	Factor6	Factor7
Factor1	**0.779**						
Factor2	0.576	**0.838**					
Factor3	0.462	0.397	**0.745**				
Factor4	0.427	0.540	0.508	**0.750**			
Factor5	0.452	0.517	0.584	0.613	**0.752**		
Factor6	0.212	0.430	0.390	0.506	0.567	**0.766**	
Factor7	0.444	0.451	0.475	0.436	0.554	0.392	**0.813**

The square roots of AVE are bold in diagonal

Table 4 Descriptive results of the digital research literacy

	Level		Needs	
Dimension	Mean	SD	Mean	SD
Digital Research Awareness	4.46	0.540	4.36	0.698
Digital Research Knowledge	3.91	0.771	4.39	0.689
Literature Skills	4.29	0.612	4.52	0.625
Auxiliary Tool Usage	3.82	0.786	4.48	0.655
Research Data Processing	4.07	0.698	4.53	0.650
Academic Communication	3.31	0.893	4.44	0.686
Digital Research Ethics	4.35	0.610	4.19	0.892
Overall	3.98	0.528	-	-

Table 5 Descriptive results for current and desired promotion channels

	Current Channels		Desired Channels		
Item	Mean	SD	Mean	SD	Differences
C1 Autonomous learning through online courses, internet resources, etc.	4.32	0.693	4.34	0.773	0.02
C2 Self-exploration of digital tools.	4.29	0.703	4.21	0.905	−0.08
C3 Communication with team members or peers.	4.20	0.756	4.28	0.778	0.08
C4 Guidance from supervisors.	3.71	1.033	4.01	0.984	0.30
C5 Relevant courses or training offered by the university.	3.40	1.144	4.39	0.721	**0.99**
C6 Activities organized by external institutions.	3.05	1.203	3.91	1.035	**0.86**
C7 Involvement in actual research projects.	4.10	0.824	4.40	0.704	0.30
C8 Participating in specialized digital skills competitions.	3.00	1.291	3.70	1.172	0.70

research literacy ranged between 3.31 and 4.46, with an overall mean of 3.98 (SD = 0.528). All mean values of students' desired training content were greater than 4, ranging between 4.19 and 4.53.

Table 5 presents the descriptive results for current and desired promotion channels, as well as the differences between them. The mean values of the scores for students' current promotion channels ranged between 3.00 and 4.32. The mean values of the scores for students' desired promotion channels ranged between 3.70 and 4.40. Their differences ranged between −0.08 and 0.99. The desired channel values were higher than the current channel values, except for C2.

5.4 Differential analysis

The specific variables exhibiting significant differences between the male and female groups are presented in Table 6. In terms of desired promotion channels, the mean values for the female group were higher for four channels. In terms of desired training content, the mean values for the female group were higher for three dimensions.

The specific variables exhibiting significant differences between the master's and doctoral groups are presented in Table 7. In terms of digital research literacy, the mean values for the doctoral group were higher, both overall and in five specific dimensions. In terms of current promotion channels, the doctoral group scored one

Table 6 Independent T-test for gender differences in different variables

	Item	Gender	Mean	SD	T	*P*
Desired promotion channels	C1	M	4.26	0.847	−2.49	0.013
		F	4.47	0.628		
	C2	M	4.10	0.982	−2.77	0.006
		F	4.38	0.747		
	C4	M	3.91	1.051	−2.09	0.037
		F	4.15	0.859		
	C7	M	4.34	0.738	−2.00	0.047
		F	4.49	0.642		
Desired training content	Literature Skills	M	4.45	0.648	−2.27	0.024
		F	4.61	0.579		
	Research Data Processing	M	4.47	0.689	−1.96	0.048
		F	4.61	0.579		
	Digital Research Ethics	M	4.11	0.961	−2.01	0.045
		F	4.32	0.766		

M, Male group; *F,* Female group

Table 7 Independent T-test for degree differences in different variables

	Item	Degree	Mean	SD	T	*P*
Digital Research Literacy	Overall	Master	3.92	0.525	−3.439	0.001
		Doctor	4.13	0.511		
	Digital Research Knowledge	Master	3.84	0.794	−2.412	0.016
		Doctor	4.06	0.706		
	Literature Skills	Master	4.23	0.619	−2.650	0.008
		Doctor	4.42	0.581		
	Auxiliary Tool Usage	Master	3.76	0.811	−2.473	0.014
		Doctor	3.98	0.716		
	Research Data Processing	Master	3.99	0.720	−2.933	0.004
		Doctor	4.23	0.626		
	Academic Communication	Master	3.19	0.854	−2.992	0.003
		Doctor	3.51	0.932		
Current promotion channels	C7	Master	4.01	0.855	−2.766	0.006
		Doctor	4.28	0.734		
Desired promotion channels	C4	Master	4.12	0.876	2.478	0.014
		Doctor	3.81	1.139		

item higher (C7). In terms of desired promotion channels, the master's group scored one item higher (C4).

The specific variables exhibiting significant differences between the nature sciences and engineering (NS&E) group and the humanities and social sciences (H&SS) group are presented in Table 8. In terms of digital research literacy, the mean values for the NS&E group were higher, both overall and in five specific dimensions. In terms of current promotion channels and desired training content, the mean values for the NS&E group were higher for three and two items, respectively.

Table 8 Independent T-test for discipline differences in different variables

	Item	Discipline	Mean	SD	T	*P*
Digital Research Literacy	Overall	NS&E	4.02	0.536	2.723	0.007
		H&SS	3.82	0.472		
	Digital Research Awareness	NS&E	4.50	0.534	2.273	0.024
		H&SS	4.32	0.547		
	Digital Research Knowledge	NS&E	3.99	0.755	3.514	0.001
		H&SS	3.62	0.766		
	Literature Skills	NS&E	4.33	0.613	2.456	0.015
		H&SS	4.13	0.584		
	Auxiliary Tool Usage	NS&E	3.89	0.808	2.636	0.009
		H&SS	3.60	0.654		
	Research Data Processing	NS&E	4.12	0.671	2.249	0.025
		H&SS	3.90	0.774		
Current promotion channels	C2	NS&E	4.34	0.699	2.331	0.022
		H&SS	4.11	0.693		
	C7	NS&E	4.15	0.807	2.140	0.033
		H&SS	3.91	0.868		
	C8	NS&E	3.08	1.295	2.072	0.039
		H&SS	2.70	1.243		
Desired training content	Digital Research Awareness	NS&E	4.41	0.684	2.435	0.015
		H&SS	4.17	0.725		
	Digital Research Knowledge	NS&E	4.44	0.675	2.641	0.009
		H&SS	4.19	0.710		

NS&E, Natural sciences and engineering group; *H&SS,* humanities and social sciences group

6 Discussion

6.1 The digital research literacy level

The descriptive results indicate that the overall level of digital research literacy among graduate students was relatively good. The means of four of the seven dimensions were above 4.0. This indicates that the majority of students had a strong awareness of using digital technologies and tools to enhance their research effectiveness, were confident in their ability to adhere to academic ethical norms in the digital environment, and possessed good literature and data processing skills applicable to the digital environment.

Although the graduate students had a strong awareness of digital research, their knowledge of the specifics of digital research was relatively limited, indicating that there is still considerable room for improvement in this dimension. The participants often did not fully recognize the impact of digital technology on their professional knowledge, and they had an insufficient understanding of the digital tools and technologies necessary for the management and implementation of research projects, which is an important reason for their weak overall digital research knowledge. The participants also exhibited some limitations in using digital tools to assist in research design and practice. Integrating digital technology into the research design

and implementation stages poses significant challenges for the graduate students and is thus an important area for improvement.

The graduate students were least proficient in using digital technology for academic presentation and communication. The survey results show that their performance in terms of academic communication skills was significantly lower than in other dimensions. Specifically, they were not good at using digital tools for online academic communication and information sharing, much as was observed by García Robelo and Veytia Bucheli (2018). The students lack the habit of sharing their research processes, experiences, and achievements on online platforms, as well as the habit of engaging in science popularization on social media.

In summary, the graduate students evaluated their own digital research literacy highly, but there was also room for further improvement.

6.2 Needs for training content

Studies to date have seldom inquired into the needs of the respondents. The current study thus investigated the students' needs regarding training content and improvement channels. The graduate students voiced a strong interest in improving digital research literacy across all dimensions and highlighted comprehensive training needs. Specifically, the average score for all dimensions was above 4, with the strongest demand for literature skills and data information processing skills, both above 4.5. Studies to date have revealed students' weaknesses in these areas (García Robelo & Veytia Bucheli, 2018; Ipanaqué-Zapata et al., 2023). It is evident that although the students have a decent self-evaluation of their digital research literacy, they also have a strong desire to further enhance it.

6.3 Current and desired promotion channels

The survey results indicated that the two main methods for the development of graduate students' digital research literacy were self-learning through online resources and self-exploration of digital tools, consistent with the findings of Bell (2021). This suggests that the graduate students have a high degree of self-reliance in terms of enhancing their digital research literacy. Peer communication and practical participation were also significant pathways for improvement.

In addition to the current main channels mentioned above, students voiced a strong desire for the university to offer relevant courses or training to enhance their digital research literacy. The gap between demand and current offerings is most pronounced in this area, and thus the university should enhance its provision of relevant courses and training activities.

6.4 Group differences

No gender differences were observed in the scores on the overall digital research literacy scale and its seven dimensions. However, the survey results indicate that females expressed a stronger desire for improvement in both content and channel

needs compared to males. This could be associated with the general tendency of females to be more receptive to learning.

The doctoral students outperformed master's students in overall digital research literacy and in five of its dimensions. This outcome aligns with expectations, since doctoral students, particularly those in advanced years, have undergone more extensive research training and frequently utilize digital technologies and tools, which contributes to their higher level of digital research literacy. Furthermore, as for current methods to improve digital research literacy, doctoral students tend to enhance their skills through direct involvement in research activities. In terms of desired promotion channels, the master's students expressed a stronger preference for improving their digital research literacy through supervision from their advisors. These disparities correlate with the varying research experiences and levels of autonomy among master's and doctoral students.

As for differences between disciplines, students in the natural sciences and engineering exhibited higher performance than those in the humanities and social sciences in terms of overall digital research literacy, as well as in five specific dimensions. A Portuguese study revealed that professors at polytechnic universities had a significantly higher level of digital competence in the area of Empowering Learners (EL) compared to their counterparts at comprehensive universities (Santos et al., 2021). Although this study focused on professors, not students, it highlights the disparity in digital literacy across disciplines. Regarding the current pathways for advancement, students in the natural sciences and engineering more frequently boosted their digital literacy through self-directed exploration, practical participation, and involvement in related competitions. In terms of their needs, these students expressed a stronger inclination to enhance their awareness and understanding of digital research.

In summary, there are many differences in the level of digital research literacy, current improvement channels, and improvement needs among different groups of students.

7 Suggestions

Graduate students voiced a strong desire to improve their digital research literacy. We recommend establishing a targeted and systematic training system to further enhance the digital research literacy of graduate students, which will support the individual development of students and the academic advancement of universities.

Provide a richer array of independent learning resources Independent learning is a primary pathway for students to enhance their digital research literacy. Universities, particularly their library departments, can establish digital research resource-sharing platforms and create subject-specific learning resource libraries to help students systematically improve their digital research literacy. By integrating relevant learning resources from both on and off campus, such as tutorials, guidelines, and research cases, universities can effectively support graduate students in their independent learning (Bell, 2021).

Enrich the supply of courses and training Students have a high demand for relevant courses and training. Universities can respond by offering specialized courses in digital research literacy to enhance their students' skills (Bell, 2021; Ipanaqué-Zapata et al., 2023; Leshchenko et al., 2021). They should also focus on integrating digital research literacy into other courses (García Robelo & Veytia Bucheli, 2018; Ipanaqué-Zapata et al., 2023; Indah et al., 2022). In addition, universities can employ internal and external resources to regularly organize lectures, training sessions, seminars, and workshops related to digital research literacy (Leshchenko et al., 2021), thereby providing graduate students with the latest developments and technical support and stimulating their enthusiasm for conducting research using digital technologies.

Conduct targeted training based on group characteristics We found significant differences in digital research literacy among graduate students in different groups. These differences must be taken into consideration when developing appropriate strategies for students with different academic backgrounds and at different stages. Targeted education and training in digital research literacy should be provided (Oliinyk et al., 2023). By offering courses at different levels ranging from basic to advanced, as well as more cutting-edge and professional seminars, the needs of different students can be met (Leshchenko et al., 2021).

Provide targeted guidance in actual research Supervisors can develop digital research literacy improvement plans for their students based on their actual research projects (Ipanaqué-Zapata et al., 2023). They can provide necessary guidance and encourage students to participate in specific digital research skills training. This guidance does not need to be overly detailed but should offer a learning framework and direction, enabling graduate students to delve deeper into self-study. Through active participation, graduate students can apply their acquired digital skills in practice, which will help them better understand and master these skills.

Strengthen the cultivation of online academic communication skills Guidance should be provided to help graduate students actively participate in online academic communication, make full use of various academic platforms, and share the latest research findings and academic viewpoints. Graduate students often enhance their digital research literacy more through peer communication than through direct supervision from their advisors. They can also join online communities to exchange experiences and resources related to digital research with other graduate students (García Robelo & Veytia Bucheli, 2018; Romaniuk et al., 2024).

Establish a university-level working team A working team should be responsible for the strategic planning and support of the university's digital research literacy initiatives. This includes continuously improving the university's digital research environment, coordinating the development of relevant learning resources across campus, designing and offering related courses and training, and establishing a digital research literacy counseling center to provide personalized guidance and address questions from teachers and students (Bell, 2021).

8 Conclusions and future works

The swift evolution of digital technology has catalyzed a paradigm shift in scientific research methodologies, concurrently elevating the demand for heightened digital literacy among researchers. As pivotal contributors to the scientific community, the enhancement of postgraduate students' digital research literacy is of paramount importance.

Postgraduate students generally rated their digital research literacy favorably, yet acknowledged areas necessitating further refinement. Their self-assessment revealed relative weaknesses in academic communication skills, utilization of research auxiliary tools, and digital research knowledge. Simultaneously, they exhibited a robust appetite for advancing their digital research literacy across all facets, articulating a comprehensive spectrum of training needs. Notably, the desire for improvement was most pronounced in literature skills and data information processing capabilities. These insights offer valuable guidance for the formulation of subsequent training initiatives. On one hand, university-led training programs should encompass all dimensions of digital research literacy, delivering a holistic educational experience. On the other hand, emphasis should be placed on enhancing literature skills, data information processing skills, leveraging digital tools for online academic communication and information sharing, embedding digital technology within research design and execution phases, and comprehending the influence of digital advancements on their specialized knowledge and research endeavors.

Self-directed learning and autonomous exploration emerged as the principal avenues through which students augmented their digital research literacy. Peer interaction and practical participation were also identified as significant pathways for development. This situation, to some extent, reflects an adaptive response to the current absence of structured training systems. While these methods remain preferred avenues for future enhancement among postgraduates, there is a concurrent, substantial demand for university-provided courses and training programs. In essence, the students expressed a pronounced need for institutional offerings that the university has yet to fully address. Consequently, it is imperative for universities to design and offer courses and training modules tailored to the digital research literacy needs of their postgraduate cohorts.

Moreover, significant variations exist among different student cohorts. For instance, doctoral students frequently augment their digital research literacy by actively engaging in research endeavors; master's students expressed a stronger preference for enhancing their digital research literacy under the mentorship of their supervisors. In terms of overall digital research literacy and its five constituent dimensions, students in the natural sciences and engineering fields outperformed their counterparts in the humanities and social sciences. Additionally, students in the natural sciences and engineering demonstrated a more pronounced aspiration to advance their awareness and understanding of digital research. Consequently, it is imperative to consider the foundational knowledge and characteristics of the target audience when developing and delivering a diverse array of courses and training programs aimed at enhancing digital research literacy.

Drawing from the survey findings, this study advocates for the establishment of a targeted and systematic training framework for digital research literacy. This framework should encompass enriching resources to bolster independent learning, developing a focused curriculum and training regimen, conducting targeted training based on group characteristics, providing guidance in actual research projects, strengthening the cultivation of online academic communication skills, and establishing a university-level digital research literacy support team.

This study has some limitations that should be noted. Primarily, the investigation was confined to postgraduate students from a single institution, and the survey sample exhibited a distinct disciplinary bias, with natural science and engineering students representing a significant majority of the total participants. Additionally, the sample size was relatively modest. These sampling constraints may affect the generalizability of the research outcomes. Future investigations should aim to broaden the sample to include a more diverse array of universities. Secondly, this study only analyzed data from scales and questionnaires. More extensive interview-based research could be undertaken in the future to substantiate the findings more robustly.

How to design courses and training programs to effectively enhance graduate students' digital research literacy, and how to integrate the cultivation of digital research literacy with professional courses and practical research processes, are important directions worthy of further exploration in the future. Additionally, the digital research landscape is in a state of perpetual flux, with digital research tools undergoing continuous innovation, and the benchmarks for postgraduate students' digital research literacy evolving accordingly. Consequently, the assessment items for digital research literacy should thus be continuously updated according to the ongoing development of digital technology, particularly in light of advancements in AI technology.

Appendix 1

Items of the Postgraduate Digital Research Literacy Scale

Item No.	Item description
Q1	I recognize the importance and value of digital technologies in research.
Q2	I am aware of using digital technologies to enhance research efficiency.
Q3	I am aware of using digital technologies to improve research quality.
Q4	I have the awareness of continuously improving my digital capabilities in a digital research environment.
Q5	I understand the impact of digital technologies on the updating of knowledge content in my field.
Q6	I understand the influence of digital technologies on the mode of knowledge production in my field.
Q7	I am familiar with the main digital technologies used in research methods in my field.
Q8	I understand the main digital technologies used in the operation and management of research projects in my field.
Q9	I am able to use search engines to search for relevant literature.

Item No.	Item description
Q10	I can retrieve relevant literature using professional academic databases.
Q11	I am proficient in the retrieval techniques of digital literature.
Q12	I can manage literature using digital tools.
Q13	I can use digital technologies to assist in research design or experimental design.
Q14	I can implement research plans using digital tools or equipment.
Q15	I can manage research-related materials using cloud backup tools.
Q16	I can perform data processing and analysis with the help of software tools.
Q17	I can present and interpret data using data visualization tools.
Q18	I can process text-based materials using digital tools.
Q19	I am skilled in text processing tools and typesetting techniques.
Q20	I can create charts and graphs required for papers using drawing tools.
Q21	I am familiar with the operation of the submission system.
Q22	I can manage my online academic identity.
Q23	I can use academic communication platforms to contact relevant scholars or research partners.
Q24	I can share research data and information with others through digital tools.
Q25	I can engage in online academic communication using digital tools.
Q26	I can conduct science popularization through social media platforms.
Q27	I adhere to research ethics when using AI technology to assist in research.
Q28	I respect and protect the digital rights of others during the research process.
Q29	I maintain a truth-seeking and pragmatic academic spirit in the digital research environment.
Q30	I maintain a rigorous and disciplined approach to scholarship in the digital research environment.

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Declarations

Conflict of interest The authors declare that they are unaware of any potential competing financial interests or personal relationships that might have appeared to influence the work described in this manuscript.

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